

Assignment-3

➤ Derive the Average & RMS value of a standard sinusoidal signal.

Let the alternating quantity be

$$x(t) = X_m \sin \omega t$$

where,

X_m = maximum (peak) value

$$\omega t = \theta$$

Average value of a sinusoidal wave:

For a Sinusoidal AC quantity, average value is taken over one half cycle (because over a full cycle it becomes zero).

Derivation:

Average value is defined as: π

$$X_{avg} = \frac{1}{\pi} \int_0^{\pi} X_m \sin \theta d\theta$$

$$= \frac{X_m}{\pi} \int_0^{\pi} \sin \theta d\theta$$

$$= \frac{X_m}{\pi} [-\cos \theta]_0^{\pi} = \frac{X_m}{\pi} [(-\cos \pi) + \cos 0]$$

$$= \frac{X_m}{\pi} [(1+1)] = X_{avg} = \frac{2X_m}{\pi}$$

$$\Rightarrow X_{avg} = 0.637 X_m$$

RMS value of a sinusoidal wave:

RMS (Root mean Square) value is the effective value of AC definition:

$$X_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} x^2(t) d\theta}$$

Substitute $x(t) = X_m \sin \theta$:

$$X_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} X_m^2 \sin^2 \theta d\theta}$$

$$= X_m \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2 \theta d\theta}$$

using identity:

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2} \quad \int_0^{2\pi} \sin^2 \theta d\theta = \pi$$

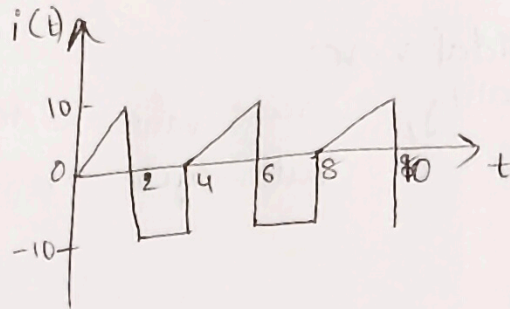
So,

$$X_{rms} = X_m \sqrt{\frac{\pi}{2\pi}}$$

$$X_{rms} = \frac{X_m}{\sqrt{2}}$$

$$X_{rms} = 0.707 X_m$$

2) Determine the average and rms value for given waveform.



From the wave form,

period $T = 4$

For $0 \leq t < 2$: $i(t) = 5t$

For $2 \leq t < 4$: constant value

$$i(t) = -10$$

Average value:

$$I_{avg} = \frac{1}{T} \int_0^T i(t) dt$$

$$= \frac{1}{4} \left[\int_0^2 5t dt + \int_2^4 (-10) dt \right]$$

$$= \frac{1}{4} [10 - 20] = -2.5 \text{ A}$$

RMS value:

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

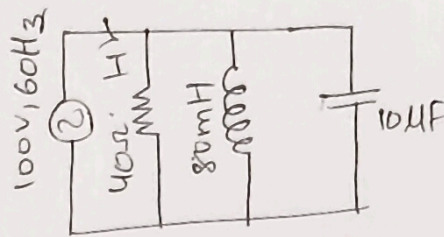
$$= \sqrt{\frac{1}{4} \left[\int_0^2 (5t)^2 dt + \int_2^4 (10)^2 dt \right]}$$

$$= \sqrt{\frac{1}{4} \left[\frac{200}{3} + 200 \right]}$$

$$= 8.16 \text{ A}$$

$$I_{rms} = 8.16 \text{ A}$$

3) Determine Admittance, impedance, phase angle, current, active power, reactive power and apparent power.



given,

$$V = 100V, f = 60\text{Hz}$$

$$R = 40\Omega$$

$$L = 80\text{mH}$$

$$C = 10\mu\text{F}$$

$$X_L = 2\pi fL = 30.16\Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times \pi \times 60 \times 10 \times 10^{-6}} = 265.25\Omega$$

$$Y_R = \frac{1}{R} = \frac{1}{40} = 0.025\text{S}$$

$$Z_L = jX_L = j30.16$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{30.16j} = -j\frac{1}{30.16}$$

$$Y_C = -0.033\text{jS}$$

$$Z_C = -jX_C = -j265.25$$

$$Y_C = \frac{1}{Z_C} = -\frac{1}{j265.25}$$

$$= 0.0037\text{jS}$$

$$\text{Total Admittance: } Y = 0.025 - j0.033 + j0.0037$$

$$= 0.025 - j0.0293 \quad |Y| = 0.0382\text{S}$$

$$|Y| = 0.038\text{S}$$

$$\theta = \tan^{-1} \left(\frac{-0.0293}{0.025} \right)$$

$$= -49.6^\circ$$

$$Y = 0.0382 \angle -49.6^\circ\text{S}$$

$$Z = \frac{1}{Y}$$

$$|Z| = \frac{1}{0.0382} = 26.18\Omega$$

$$\angle Z = 49.6$$

$$Z = 26.18 \angle 49.6^\circ\Omega$$

$$I = \frac{V}{Z}$$

$$I = \frac{100}{26.18} = 3.819\text{Amps}$$

$$I = 3.819 \angle -49.6$$

$$\cos(49.6) = \cos\phi = 0.65$$

$$P = VI \cos\phi$$

$$= 100 \times 3.819 \times 0.65 = 248.2\text{W}$$

$$Q = VI \sin \phi$$

$$= 100 \times 3.819 \times \sin 49.6$$

$$= 100 \times 3.819 \times 0.76$$

$$= 290.4 \text{ VAR}$$

$$S = VI$$

$$= 100 \times 3.819$$

$$= 381.9 \text{ VA}$$

4) Derive the expressions for half power frequencies in a series RLC resonance circuit.

given

resistance = R , Inductance = L , capacitance = C .

Angular frequency = ω .

$$\text{Impedance, } Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\text{current, } I = \frac{V}{Z}$$

$$\text{power dissipated, } P = I^2 R = \frac{V^2 R}{Z^2}$$

for maximum power,

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Impedance at resonance: $Z = R$, Maximum power $P_{\max} = \frac{V^2}{R}$

Half power condition,

Half-power frequencies ω_1, ω_2 are defined as frequencies at which power is half of maximum power.

$$P = \frac{P_{\max}}{2}$$

$$\frac{V^2 R}{Z^2} = \frac{1}{2} \left(\frac{V^2}{R} \right)$$

cancel V^2 :

$$\frac{R}{Z^2} = \frac{1}{2R} \Rightarrow Z^2 = 2R^2 \Rightarrow Z = \sqrt{2} R$$

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2} R$$

$$R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$$

$$\omega L - \frac{1}{\omega C} = \pm R$$

$$\omega^2 LC - 1 = \pm \omega RC$$

$$\omega^2 LC \pm \omega RC - 1 = 0$$

It is a quadratic in ω ,

$$\omega_1, \omega_2 = \frac{RC \pm \sqrt{(RC)^2 - 4(LC)(-1)}}{2LC}$$
$$= \frac{RC}{2LC} \pm \sqrt{\left(\frac{RC}{2LC}\right)^2 + \frac{4LC}{(2LC)^2}}$$

$$\omega_1, \omega_2 = \frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \text{ rad/sec.}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \text{ rad/sec}$$

frequency should be positive.

we can relate the half power frequency with resonance frequency.

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$